

STAT 380.201

Finals Part II - Data Science Project

Predicting Online Purchase Intention
and Discovering Shopper Segments

Due: 11 : 59 pm ET, Aug 10, 2026
Total: 100 points

Project question

An e-commerce company wants to understand two related questions:

1. Can browsing behavior be used to predict whether a browsing session will end in a purchase?
2. If the purchase outcome is hidden, are there distinct behavioral segments among online shopping sessions?

You will answer the first question with supervised classification and the second with unsupervised K-means clustering.

Student name : _____

Submission date: _____

May your code run, your files upload, and your internet remain stable.

Project overview

The project integrates the major ideas developed in this course: exploratory data analysis, categorical-variable encoding, training/validation/test separation, logistic regression, K-nearest-neighbor classification, cross-validation, confusion-matrix metrics, ROC/AUC analysis, and K-means clustering.

Required submissions

Submit the following two files:

1. a polished PDF report, normally 8-10 pages excluding references and a code appendix; and
2. one reproducible R script or R Markdown file, or Quarto file that generates every reported number, table, and figure.

Reproducibility and test-set rule

Use `set.seed(380)` before all random operations unless a question gives another seed. The test set must remain untouched until you have selected the final model and fixed all tuning choices. Looking at test performance while selecting a model is data leakage.

“ With great final project comes great responsibility . . . to submit the correct final files!” - Anonymous

Dataset and data access

Source and context

Use the *Online Shoppers Purchasing Intention Dataset* from the UCI Machine Learning Repository. It contains 12,330 distinct online shopping sessions collected over one year. The response Revenue records whether a session ended in a transaction. The official dataset description reports 10,422 non-purchase sessions and 1,908 purchase sessions, so the positive class is substantially less common than the negative class.

Official source

Dataset webpage: <https://archive.ics.uci.edu/dataset/468/online+shoppers+purchasing+intention+dataset>

Direct CSV: https://archive.ics.uci.edu/ml/machine-learning-databases/00468/online_shoppers_intention.csv

A minimal import command is provided only to standardize access.

```
shop <- read.csv(
  "https://archive.ics.uci.edu/ml/machine-learning-databases/00468/online_shoppers_intention.csv",
  stringsAsFactors = FALSE
)
```

Variable dictionary

The integer codes for operating system, browser, region, and traffic type are category labels, not quantitative measurements. Treat them as factors for supervised modeling.

Variable	Variable type	Meaning
Administrative	Quantitative	Number of administrative/account-management pages visited in the session.
Administrative_Duration	Quantitative	Total time spent on administrative pages.
Informational	Quantitative	Number of informational pages visited.
Informational_Duration	Quantitative	Total time spent on informational pages.
ProductRelated	Quantitative	Number of product-related pages visited.
ProductRelated_Duration	Quantitative	Total time spent on product-related pages.
BounceRates	Quantitative	Bounce-rate metric associated with pages visited during the session.
ExitRates	Quantitative	Exit-rate metric associated with pages visited during the session.
PageValues	Quantitative	Page-value metric for pages visited in the session.
SpecialDay	Quantitative	Closeness of the visit date to a special shopping day; values range from 0 to 1.
Month	Categorical	Month in which the session occurred.
OperatingSystems	Categorical	Coded operating-system category.
Browser	Categorical	Coded browser category.
Region	Categorical	Coded geographic-region category.
TrafficType	Categorical	Coded source/channel through which the visitor arrived.
VisitorType	Categorical	Visitor category, such as returning, new, or other.
Weekend	Categorical	Whether the session occurred on a weekend.
Revenue	Binary response	TRUE if the session ended in a transaction and FALSE otherwise.

Positive class

Throughout the project, define **Purchase** (Revenue = TRUE) as the positive class. State the names/descriptions of rows and columns whenever you display a confusion matrix.

Grading map

Section	Points
A. Data understanding and exploratory analysis	24
B. Data splitting and preprocessing	14
C. Logistic regression	22
D. K-nearest neighbors and 10-fold cross-validation	16
E. Response-free K-means clustering	18
F. Communication and reproducibility	6
Total	100

Project questions

A. Data understanding and exploratory analysis

(24 points)

A1. Import, structure, and data quality.

[4 points]

Import the data and report its dimensions. Examine variable classes, missing values, duplicated rows, impossible values, and unusual coding. Convert the seven categorical predictors and the response to suitable R types. Clearly document every data-cleaning or recoding decision. Do not silently delete observations.

A2. Class balance and the naive benchmark.

[4 points]

Create a count and proportion table for Revenue and an appropriate visualization. Construct the naive classifier that predicts the majority class for every session. Report its accuracy and explain why a high accuracy can be misleading for this dataset.

A3. Exploratory visual analysis.

[16 points]

Produce a coherent set of exploratory summaries and figures that includes, at minimum:

- distributions of at least four quantitative variables;
- purchase rates for at least three categorical predictors;
- at least three quantitative-predictor comparisons by purchase outcome; and
- a correlation display for the genuinely quantitative predictors.

Every plot must have an informative title, labeled axes, units where applicable, and a brief interpretation. Avoid presenting many nearly identical plots without a clear analytical purpose.

B. Data splitting and preprocessing

(14 points)

B1. Training, validation, and test sets.

[8 points]

Using `set.seed(380)`, randomly divide the observations into approximately 70% training, 30% test sets. Preserve the purchase proportion reasonably well in all datasets. Report the dimensions and purchase proportions of each set, and verify that no observation belongs to more than one set.

B2. Method-specific preprocessing.

[6 points]

Prepare the data separately for the following model families:

- **Logistic regression:** use meaningful factor variables and consistent reference levels.
- **K-nearest neighbors:** construct a numeric design matrix using dummy/one-hot encoding and standardize predictors. Estimate centering and scaling constants from the training data only, then apply the same constants and column structure to validation and test data.

Explain why the coded values of Browser, Region, and similar variables must not be treated as ordinary numerical distances.

C. Logistic regression

(22 points)

C1. Fit a main-effects logistic model.

[8 points]

On the training data, fit a logistic regression model for the probability of purchase using all 17 predictors as main effects. State the fitted model formula, the positive class, and the baseline level used for each categorical predictor. Report a compact coefficient table rather than pasting the entire console output.

C2. Interpret coefficients in context.**[6 points]**

Interpret at least three fitted effects using odds ratios:

- one quantitative predictor using a practically meaningful increase rather than automatically using one unit;
- one binary predictor; and
- one level of a multi-level categorical predictor relative to its baseline.

Use “holding all other included predictors constant” language .

C3. Validation ROC curve, threshold, and confusion metrics.**[6 points]**

Generate predicted purchase probabilities for the validation set. Report and interpret the ROC curve and AUC. Evaluate both the default threshold 0.50 and a threshold chosen from the validation ROC curve using the point closest to the top-left corner. For each threshold, report a confusion matrix, accuracy, sensitivity, specificity, precision, and misclassification rate. Choose one threshold and justify it using the relative consequences of false positives and false negatives in a stated business use case.

C4. Logistic-model limitations.**[2 points]**

Identify two limitations of the fitted logistic model. Possibilities include class imbalance, nonlinearity, omitted interactions, correlated predictors, unstable interpretation, or unavailable predictors at deployment time. Explain why these limitations matter.

D. K-nearest neighbors and 10-fold cross-validation**(16 points)****D1. Select the number of neighbors by 10-fold CV.****[10 points]**

Using only the training data and the standardized numeric design matrix, perform 10-fold cross-validation for odd values of k from 1 through 31. For every candidate k , compute the mean cross-validated misclassification rate, sensitivity, specificity, and balanced accuracy. Plot at least one cross-validation criterion against k , select a value of k , and explain how you handled a near-tie.

D2. Validate the selected KNN classifier.**[4 points]**

Fit the selected KNN classifier using the full training set and evaluate it on the validation set. Report its confusion matrix and the same threshold-based metrics used for logistic regression. Compare KNN to your chosen logistic-regression threshold.

D3. Explain the role of scaling and k .**[2 points]**

Explain why standardization is essential for a distance-based method. Then describe the bias-variance behavior of very small and very large values of k , and connect this explanation to the shape of your cross-validation plot.

E. K-means clustering**(18 points)****Strict unsupervised rule**

Create a new data object with Revenue removed before any clustering choice is made. Do not use purchase labels to select variables, transformations, K , initializations, or clusters. You may reattach Revenue only after the final cluster assignments are fixed, for post-hoc interpretation in E4.

E1. Construct the clustering feature matrix.**[4 points]**

Use the ten genuinely quantitative predictors listed in the variable dictionary. Exclude the integer-coded categorical predictors because their numerical codes do not define meaningful Euclidean distances. Standardize all clustering variables. Revenue.

Diagnostic for choosing K

Elbow diagnostic: For each K , record the best total within-cluster sum of squares from multiple initializations. Look for a point after which additional clusters produce much smaller improvements.

E2. Select the number of clusters.**[6 points]**

For $K = 1, \dots, 10$, run K-means with at least $nstart = 25$ and plot total within-cluster sum of squares against K . Compute and plot total within-cluster sum of squares for $K = 2, \dots, 10$. Select a final value of K . Explain your choice of K .

E3. Fit and profile the final clustering.**[5 points]**

Run K-means on the full standardized quantitative dataset with your selected K , $nstart = 50$, and a fixed seed. Report cluster sizes, total within-cluster variation. Create at least two informative two-dimensional plots colored by cluster. Give each cluster a concise behavioral name and justify it from its profile rather than from the purchase outcome.

E4. Post-hoc external interpretation and limitations.**[3 points]**

Only after E3 is complete, reattach Revenue and report the purchase rate within each cluster. Discuss whether the unsupervised segments happen to differ in purchase behavior, while emphasizing that Revenue did not create the clusters. State at least two limitations of K-means for these data, such as skewness, outliers, Euclidean geometry, roughly spherical-cluster preference, categorical-variable exclusion, sensitivity to initialization, or uncertainty in K .

F. Communication and reproducibility**(6 points)****F1. Executive summary.****[2 points]**

Begin the report with a 150-250 word introduction containing the dataset description, main objectives. Answer every question section-by-section. Interpret every result.

F2. Reproducibility.**[2 points]**

The submitted source file must run from a clean R session, set all random seeds, load required packages, import the data, and reproduce the report. Use clear object names and concise comments. Do not include unexplained manual edits to tables or figures.

F3. Professional presentation.**[2 points]**

Organize the report around questions and conclusions rather than providing code snippets. Number and caption tables and figures, round sensibly, define the positive class. Place any lengthy code or secondary output in an appendix.

Dataset citation

Sakar, C. and Kastro, Y. (2018). *Online Shoppers Purchasing Intention Dataset*. UCI Machine Learning Repository. DOI: <https://doi.org/10.24432/C5F88Q>. The dataset is distributed under the Creative Commons Attribution 4.0 International license.

A strong submission does more than run algorithms.

It makes defensible choices, evaluates predictions honestly, interprets results in context, and explains what the data cannot establish.

